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MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Durable and Portable. #1 (HIGH Priority)

A high priority for our group is making the plant care system durable and portable. The system needs to be strong enough to last a long time without breaking and light enough to be moved around the classroom if needed. If the system is too fragile, it could break easily, causing water spills or damage to the plants. Portability is also important because we might need to move the plants to different locations for better lighting or easier care. Using sturdy materials and a lightweight design will help keep the system functional and easy to manage over time.

Water Supply #2 (HIGH Priority)

A high priority in our group is to store and apply water effectively. We need a controlled water supply to ensure that plants receive the correct amount of water, avoiding both overwatering and underwatering. If the water supply is not reliable or properly managed, the plants could either become waterlogged from excess water or dehydrated due to insufficient watering. This can cause stress to the plants, affect their growth, and even lead to plant failure. Ensuring that the system has a consistent and efficient way to store and deliver water is essential for maintaining the plants' health and preventing any issues that could arise from improper watering.

Soil #3 (HIGH Priority)

A high priority for our group is maintaining nutrient-rich soil for the plants. Soil provides the essential nutrients that plants need to grow strong and healthy. If the soil lacks nutrients, plants may become weak, grow slowly, or even die. To prevent this, we need to

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regularly check and improve the soil by adding fertilizer, compost, or other organic matter. Making sure the soil stays healthy and full of nutrients will help the plants develop strong roots, produce healthy leaves, and grow properly in the classroom environment.

Proper Lighting #4 (High Priority)

A mid-priority for our group is to provide proper lighting for the plants. While water is the most important factor, plants also need enough light to grow strong and healthy. If they don't get enough light, they might become weak or grow slowly, and too much light could dry them out. Since classrooms have different lighting conditions, we may need to move the plants around to find the best spots, like near windows or under grow lights. Having the right amount of light will help the plants stay healthy and grow properly, but it is something we can adjust over time if needed.

Realistic size #5(High priority)

A high priority for our group is making sure the design is realistic and practical. The system should be easy to build, use, and maintain with the materials and resources we have available. If the design is too complicated or requires expensive or hard-to-find materials, it might not work well in a classroom setting. By keeping the design simple, effective, and achievable, we can make sure it functions properly and can be used by anyone caring for the plants. A realistic design helps ensure the system is reliable and sustainable in the long run.

Location stays close to win the dow #6 (Mid Priority)

A Mid-priority is where the plant is located because it could affect the growth and how our design works. We require the plant to get enough sunlight in order to grow and prosper. If the plant is not getting enough sunlight that could cause it to die and not perform its normal cycles like photosynthesis. The design requirement is a good spot near the window



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facing the sun for guaranteed energy going to the plant. The plant doesn't need too much energy because it could become brown or bleached, or the excess light could damage the chlorophyll in the leaves that could lead to stunted growth.

Aesthetic #7 (LOW Priority)

A low priority for our group is aesthetics, or how the plant setup looks in the classroom. While it's nice for the system to be organized and visually appealing, it doesn't directly affect plant health or growth. Our main focus should be on making sure the plants get enough water, light, and proper care. Even if the setup looks messy or not perfect, as long as the plants are healthy and growing well.

Stakeholder consult :

When consulting our stakeholder, Mrs. Greer, we learned that preventing mold growth is one of her top concerns. Our highest priority should be ensuring the plants stay alive. Additionally, while water conservation is a lower priority, it remains important to her. She also emphasized that the plants must stay near a window.

